



Newington College

2009

TRIAL HSC EXAMINATION

# Mathematics

## General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is proved at the back of this paper
- All necessary working should be shown in every question

## Total marks: 120

- Attempt Questions 1–10
- All questions are of equal value

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x, \quad x > 0$

Total Marks - 120

Attempt Questions 1 – 10.

All questions are equal value.

Answer each question in a SEPARATE writing booklet.

Question 1 (12 Marks) Use a SEPARATE writing booklet.

Marks

(a) Find the value of  $\sqrt{\frac{50}{\pi}}$  correct to three significant figures. 2

(b) Factorise  $3x^2 + 5x - 12$ . 2

(c) Simplify  $\frac{3}{5} - \frac{x-1}{4}$ . 2

(d) Find the primitive of  $4 - \frac{1}{x^2}$ . 2

(e) Find the values of  $x$  for which  $|x-4| > 3$ . 2

(f) If  $f(x) = \begin{cases} 7-3x & \text{for } x \leq 1 \\ x^3+1 & \text{for } x > 1 \end{cases}$

evaluate  $f(0) + f(2)$ . 2

Question 2 (12 Marks) Use a SEPARATE writing booklet.

(a) Differentiate the following:

(i)  $(2x^3 - 1)^5$  2

(ii)  $e^{5x} \tan x$  2

(iii)  $\frac{\ln x}{x^2}$  2

(b) Find the equation of the tangent to the curve  $y = 2x(1-x^2)$  at the point  $(1, 0)$ .

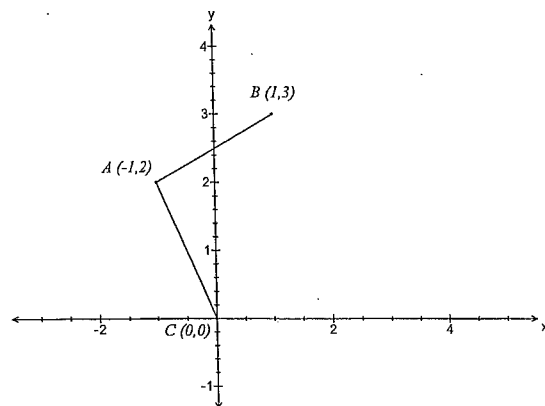
3

(c) Evaluate  $\int_2^3 \frac{3x}{x^2-1} dx$ , give answer in exact form. 3

Question 3 (12 Marks) Use a SEPARATE writing booklet

Marks

(a)



In the diagram above  $A(-1,2)$ ,  $B(2,3)$  and  $C(0,0)$  are as shown.

- (i) Prove that  $AB \perp AC$ . 2
- (ii) Find the equation of CD such that  $CD \parallel AB$ . 2
- (iii) If  $BD$  parallel to the x-axis find the co-ordinates of  $D$ . 2
- (iv) Describe the special quadrilateral  $ABDC$  geometrically. 1
- (v) Find the area of  $ABDC$ . 3

- (b) The table shows values of a function  $f(t)$  for the five values of  $t$ .

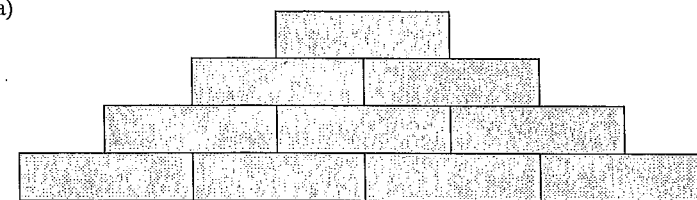
|        |   |   |   |   |    |
|--------|---|---|---|---|----|
| $t$    | 0 | 3 | 6 | 9 | 12 |
| $f(t)$ | 0 | 2 | 5 | 8 | 4  |

Use the Trapezoidal Rule with these 5 values to estimate  $\int_0^{12} f(t) dx$ . 3

Question 4 (12 Marks) Use a SEPARATE writing booklet.

Marks

(a)



Top four layers.

A bricklayer builds a triangular wall. The top of the wall is shown above. If the bricklayer uses 171 bricks, how many layers did he build?

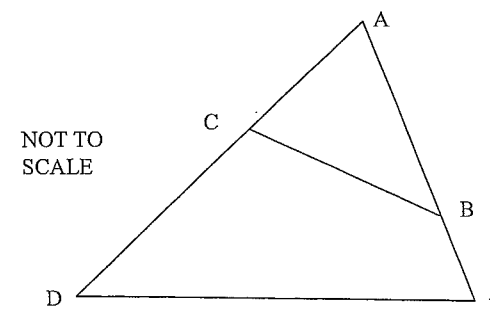
3

- (b) Sally has a bag of sweets which are all identical in shape. The bag contains 6 orange drops and 4 lemon drops. She selects one sweet at random, eats it, and then takes another at random. Determine the probability that:

- (i) both sweets were lemon drops 2
- (ii) at least one of the drops was an orange drop. 2

- (c) Evaluate  $\sum_{n=1}^{10} 2^n$ . 3

- (d) In the diagram below,  $\triangle ABC$  is similar to  $\triangle ADE$ .  $\angle ADE = \angle ABC$ ,  $AC = 8 \text{ cm}$ ,  $AB = 12 \text{ cm}$  and  $BE = 4 \text{ cm}$  Find the length of  $CD$ . 2



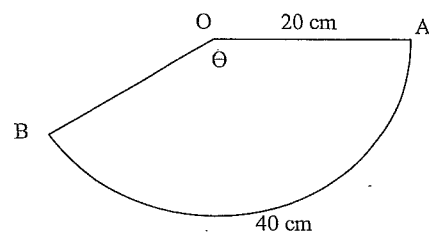
**Question 5** (12 Marks) Use a SEPARATE writing booklet.

**Marks**

(a) Solve  $\log_3(5x+1) = 2$ .

2

(b)



In the diagram  $AB$  is an arc of a circle with centre  $O$  and radius  $20$  cm. The arc has a length of  $40$  cm. Find:

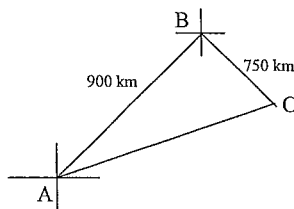
(i) the exact size of  $\angle AOB$  in radians,

1

(ii) the exact area of the sector  $AOB$ .

1

(c) A plane travels  $900$  km from  $A$  to  $B$  on a bearing of  $50^\circ T$ . The plane then travels  $750$  km on a bearing of  $135^\circ T$  to a point  $C$ .



Copy the diagram into your answer booklet.

(i) Show that  $\angle ABC$  is  $95^\circ$ , giving reasons.

2

(ii) Find the distance of  $CA$ , correct to one decimal place.

2

(iii) Find  $\angle ACB$ .

2

(iv) Find the bearing of  $A$  from  $C$ , to the nearest minute.

2

**Question 6** (12 Marks) Use a SEPARATE writing booklet.

**Marks**

(a)

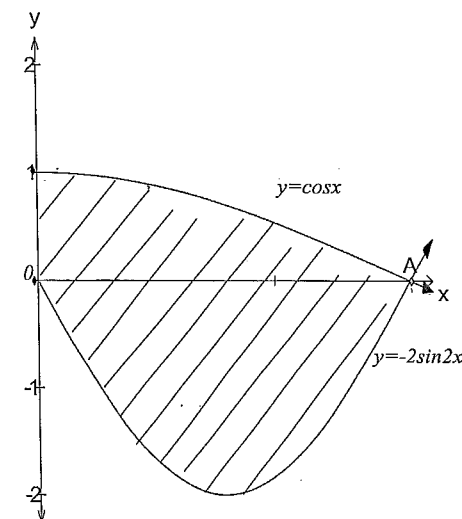


DIAGRAM NOT DRAWN TO SCALE

The diagram shows the graph of  $y = -2 \sin 2x$  and  $y = \cos x$ , for  $x \geq 0$ . The graphs intersect at point  $A$ .

(i) Show that point  $A$  has co-ordinates  $\left(\frac{\pi}{2}, 0\right)$ .

2

(ii) Find the shaded area enclosed by the two graphs for  $0 \leq x \leq \frac{\pi}{2}$ .

3

(b) Given the equation  $2x^2 + 7x - 3 = 0$  has roots  $\alpha$  and  $\beta$ , evaluate the following:

(i)  $\alpha + \beta$

1

(ii)  $\alpha \beta$

1

(iii)  $2\alpha^2 + 2\beta^2$ .

2

(c) Find the values of  $A$ ,  $B$  and  $C$ , such that:

$$4x^2 + 5x - 3 \equiv A(x-1)^2 + B(x-1) + C$$

3

**Question 7** (12 Marks) Use a SEPARATE writing booklet.

**Marks**

- (a) Given the parabolic equation:

$$8y = x^2 - 2x - 15$$

Find:

- (i) the coordinates of the vertex

2

- (ii) the coordinates of the focus

1

- (iii) the equation of the directrix.

1

- (b) For the curve  $f(x) = 2x^3 + 3x^2 - 12x - 4$

- (i) Find the coordinates of the stationary points and determine their nature.

3

- (ii) Find the coordinates of the point of inflexion.

2

- (iii) Hence sketch the graph of  $y = f(x)$ , showing the turning points, the point of inflexion and where the curve meets the  $y$ -axis.

2

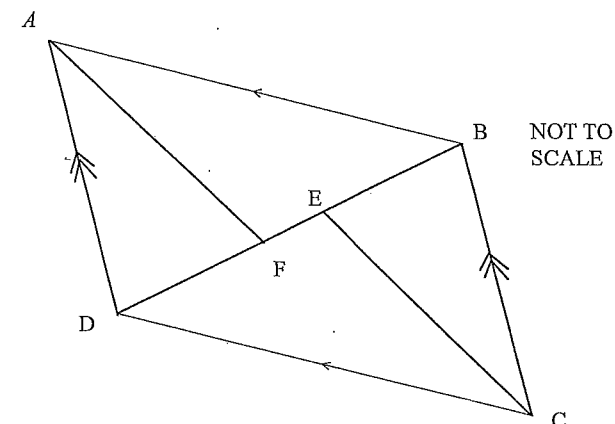
- (iv) For what values of  $x$  is the graph of  $f(x)$  concave up?

1

**Question 8** (12 Marks) Use a SEPARATE writing booklet.

**Marks**

- (a) Copy or trace the diagram into your writing booklet.



$ABCD$  is a parallelogram.  $FA$  bisects  $\angle BAD$  and  $EC$  bisects  $\angle BCD$ .

- (i) Prove that  $\triangle ADF$  and  $\triangle CBE$ .

3

- (ii) Hence find the length of  $EF$  if  $BD = 40 \text{ cm}$  and  $DF = 16 \text{ cm}$ .

1

- (b) The position  $x \text{ cm}$  of a particle  $P$  moving along the  $x$ -axis after  $t$  seconds is given by  $x = 25t - 10 \ln t \text{ cm}$ , where  $t \geq 1$ .

- (i) Find expressions for the particle's velocity and acceleration.

2

- (ii) Find the velocity and acceleration when  $t = e$  minutes.

2

- (iii) Discuss the velocity as  $t \rightarrow \infty$ .

2

- (iv) Sketch the graph of the velocity function?

2

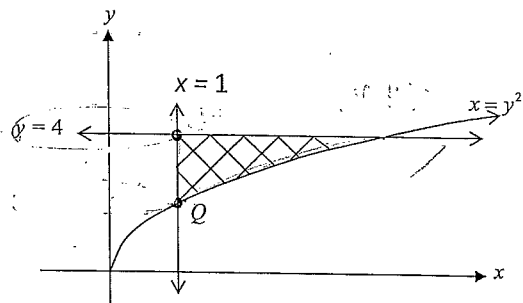
Question 9 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) Solve  $4\cos^2\theta - 3 = 0$ , if  $0 \leq \theta \leq 2\pi$ .

3

(b)



- (i) Find the coordinates of  $Q$ , the intersection of  $x = y^2$  and  $x = 1$ .
- (ii) The shaded region in the diagram is the area bounded by the lines  $y = 4$ ,  $x = 1$  and the parabola  $x = y^2$ . This region is rotated about the  $y$ -axis. Find the volume of the solid formed.

1

3

- (c) The amount  $Q$  grams of a carbon isotope in a dead tree trunk is given by  $Q = Q_0 e^{-kt}$  where  $Q_0$  and  $k$  are positive constants and time  $t$  is measured in years from the death of the tree.

- (i) Show that  $Q$  satisfies the equation  $\frac{dQ}{dt} = -kQ$

1

- (ii) Show that if the half-life of the isotope is 4500 years,  
then  $k = \frac{1}{4500} \ln 2$

2

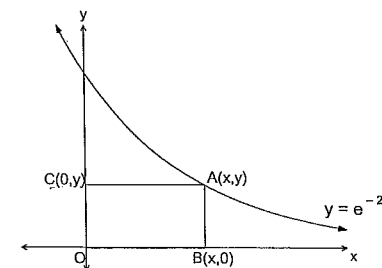
- (iii) For a particular dead tree trunk, the amount of isotope is only 15% of the original amount in the living tree. How long ago did the tree die? Give your answer to the nearest 1000 years.

2

Question 10 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) A rectangle is placed under the curve  $y = e^{-2x}$  as shown. If  $A(x)$  gives the area of  $OBAC$ ,



- (i) find an expression for  $A(x)$  in terms of  $x$ .
- (ii) determine the coordinates of  $A(x)$  such that the rectangle has a maximum area.

2

3

- (b) A couple wish to purchase a home. They need to calculate the maximum amount they can borrow given that the current interest rate is 9% pa, compounded monthly and the term of the loan is 30 years. They can afford to repay \$2700 per month, paid at the end of every month. Let  $A_n$  be the amount owing at the end of  $n$  months and  $P$  be the amount borrowed.

- (i) Show that after three months the amount of money owing is

$$A_3 = P(1.0075)^3 - 2700(1 + 1.0075 + 1.0075^2).$$

2

- (ii) Hence show that if the loan is to be paid off after  $n$  months then

$$P = \frac{360000(1.0075^n - 1)}{(1.0075)^n}$$

2

- (iii) Calculate, correct to the nearest cent, the amount that they can borrow if the loan is to be paid off in thirty years.

2

- (iv) How much interest, in dollars, will they pay for loan?

1

End of Paper.

21) (a)  $\sqrt{\frac{50}{17}} = 3.989422806 \dots$  ✓  
 $= \underline{3.99}$  [3 sig fig] ✓ 2 for correct answer only

(b)  $3x^2 + 5x - 12$  ✓  
 $= (3x - 4)(x + 3)$  ✓

(c)  $\frac{3}{5} - \frac{x-1}{6}$  ✓  
 $= \frac{12 - 5(x-1)}{20}$  ✓  
 $= \frac{12 - 5x + 5}{20}$  ✓  
 $= \frac{17 - 5x}{20}$  ✓

(d)  $\int 4 - \frac{1}{x^2} dx = \int 4 - x^{-2} dx$  ✓  
 $= 4x + x^{-1} + C$  ✓  
 $= 4x + \frac{1}{x} + C$  ✓

(e)  $|x-4| > 3$   
 $\therefore x-4 > 3$  or  $x-4 < -3$   
 $\therefore x > 7$  or  $x < 1$  ✓ ✓

(f)  $f(0) = 7 - 3(0) = 7$   
 $f(2) = 2^3 + 1 = 9$   
 $\therefore f(0) + f(2) = 7 + 9 = 16$  ✓ ✓

Q2 (a) (i)  $y = (2x^3 - 1)^5$  12

$\frac{dy}{dx} = 5(2x^3 - 1)^4 \times 6x^2$   
 $= 30x^2(2x^3 - 1)^4$  ✓ ✓

(ii)  $y = e^{5x} \tan x$   
 $\frac{dy}{dx} = e^{5x} \sec^2 x + \tan x \cdot 5e^{5x}$  ✓ ✓ Product Rule.  
 $= e^{5x} (\sec^2 x + 5 \tan x)$

(iii)  $y = \frac{\ln x}{x^2}$  Quotient Rule  
 $\frac{dy}{dx} = \frac{x^2 \times \frac{1}{x} - 2x \ln x}{x^4}$  ✓  
 $= \frac{x - 2x \ln x}{x^4}$   
 $= \frac{1 - 2 \ln x}{x^3}$  ✓

(b)  $y = 2x(1-x^2)$   
 $\therefore y = 2x - 2x^3$   
 $\frac{dy}{dx} = 2 - 6x^2$   
 at  $x=1$   $\frac{dy}{dx} = 2 - 6 = -4$  ✓  
 $\therefore$  Equation of tangent at (1, 0)  
 $y - y_1 = m(x - x_1)$  ✓  
 $y - 0 = -4(x - 1)$   
 $y = -4x + 4$  ✓  
 $\therefore 4x + y - 4 = 0$  is the equation of tangent. Note: Accepted form

Q2(c)

$$\int_2^3 \frac{3x}{x^2-1} dx$$

$$= \frac{3}{2} \int_2^3 \frac{2x}{x^2-1} dx \quad \checkmark$$

$$= \frac{3}{2} [\ln(x^2-1)]_2^3 \quad \checkmark$$

$$= \frac{3}{2} [\ln 8 - \ln 3] \quad \checkmark \quad \text{Accept either.}$$

$$= \frac{3}{2} \ln \frac{8}{3} \quad \checkmark$$

## Question 3

$$\begin{aligned} \text{a) } M_{AB} &= \frac{3-2}{1-1} & M_{AC} &= \frac{2}{-1} \\ &= \frac{1}{2} & &= -2 \end{aligned}$$

$$\text{Since } M_{AB} \times M_{AC} = -1 \quad (2)$$

$AB \perp AC$

$$\begin{aligned} \text{ii) } y &= mx \text{ as } b=0 \\ \therefore \text{Equation of } CD &\text{ is } y = \frac{1}{2}x \quad (1) \end{aligned}$$

$$\begin{aligned} \text{iii) at } D \quad y &= 3 \\ \therefore 3 &= \frac{1}{2}x \\ x &= 6 \quad (2) \\ \therefore D &= (6, 3) \end{aligned}$$

$$\text{iv) } ABCD \text{ is a Trapezium} \quad (1)$$

(Right  $\angle$ )

$$\begin{aligned} \text{v) } A &= \frac{1}{2} \times 6 \times 3 - \frac{1}{2} \times \frac{1}{2} \times 1 + \frac{1}{2} \times 2.5 \times 1 \\ &= 10 \text{ units}^2 \end{aligned}$$

(2)

$$\text{b) } \int_a^b f(x) dx = \frac{1}{2} [f(a) + f(b)]$$

$$\begin{aligned} &= \frac{3}{2} [f(0) + f(3) + f(3) + f(6) + f(6) + f(9) + f(9) + f(12)] \\ &= \frac{3}{2} [0 + 2 + 2 + 5 + 5 + 8 + 8 + 9] \\ &= 51 \end{aligned}$$

(3)

# Question 4

a)  $S_n = 171$   
 $d = 1$        $a = 1$   
 $\frac{n}{2} [2a + (n-1)d] = 171$   
 $\frac{n}{2} [2 + (n-1)] = 171$   
 $\frac{n}{2} (n+1) = 171$   
 $n(n+1) = 342$   
 $n^2 + n = 342$   
 $n^2 + n - 342 = 0$   
 $(n+19)(n-18) = 0$   
 $n = -19$        $n = 18$   
 $\therefore n = 18$  as  $n > 0$   
 $\therefore 18$  layers built (3)

b)  $P(U) = \frac{4}{10} \times \frac{3}{9}$   
 $= \frac{2}{15}$  (2)

ii)  $P(\text{At least 1 Orange}) = 1 - P(U)$   
 $= 1 - \frac{2}{15}$   
 $= \frac{13}{15}$  (3)

c)  $\sum_{n=1}^{\infty} 2^n = 2^1 + 2^2 + 2^3 + 2^4 + \dots + 2^{10}$   
 $= 2046$  (3)

OR  $= S_n \text{ GP}$   
 $= \frac{a(1-r^n)}{1-r}$   
 $= \frac{2(1-2^{10})}{1-2}$   
 $= 2046$

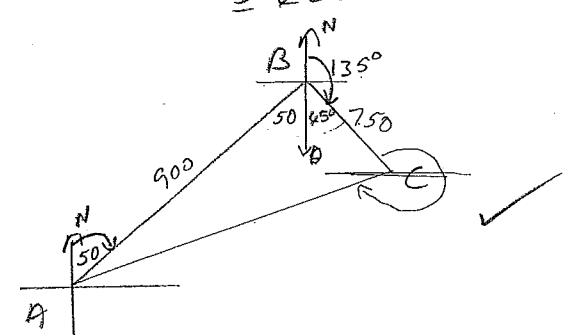
d)  $\frac{AD}{12} = \frac{16}{8}$   
 $AD = 12 \times \frac{16}{8}$   
 $AD = 24$  (2)  
 $\therefore CD = 24 - 8$   
 $= 16 \text{ cm}$   
 OR  
 $\frac{x+8}{12} = \frac{16}{8}$   
 $x+8 = 12 \times \frac{16}{8}$   
 $x+8 = 24$   
 $x = 16$   
 $\therefore DC = 16 \text{ cm}$

Q 5(a)  $\log_3(5x+1) = 2$   
 $5x+1 = 9$  ✓  
 $5x = 8$   
 $x = \frac{8}{5}$  ✓

(b) i) Arc length =  $r\theta$   
 $\therefore 20\theta = 40$   
 $\theta = 2$  radians ✓

(ii) Area of Sector =  $\frac{1}{2} r^2 \theta$   
 $= \frac{1}{2} (20)^2 \times 2$   
 $= 400 \text{ cm}^2$  ✓

(c)

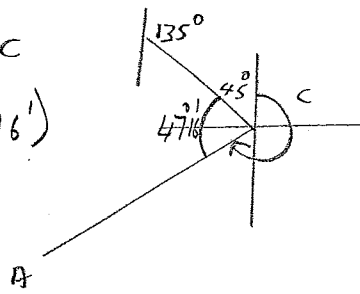


(i) From diagram  $\angle ABD = 50^\circ$  [Alternate angles]  
 $\angle DBC = 45^\circ$  [Supplementary angles] ✓  
 $\therefore \angle ABC = 95^\circ$  Note: Working can be shown on diagram.

(ii)  $CA^2 = BC^2 + AB^2 - 2 \times 750 \times 900 \times \cos 95^\circ$  ✓  
 $CA = 1220.7 \text{ km}$  [correct to one decimal place]  
 ✓ [Note: DO NOT DECREASE FOR ROUNDING]

(iii)  $\frac{\sin \angle ACB}{900} = \frac{\sin 95}{1220.7}$   
 $\sin \angle ACB = \frac{900 \times \sin 95}{1220.7}$   
 $\angle ACB = 47^\circ 16'$  [Note: Accept  $47^\circ$  or  $47.27^\circ$ ]

50) (iv) Bearing of A from C  
 $= 360^\circ - (45^\circ + 47^\circ 16')$   
 $= 267^\circ 44'$



(a) (i) If  $y = -2\sin 2x$ ,  $x = \frac{\pi}{2}$   
 $\therefore y = -2\sin \pi$   
 $= 0$   
 $\therefore$  curves intersect at  $A(\frac{\pi}{2}, 0)$

If  $y = \cos x$ ,  $x = \frac{\pi}{2}$   
 $y = \cos \frac{\pi}{2}$   
 $= 0$

(ii) Enclosed Area =  $\int_0^{\frac{\pi}{2}} [\cos x - (-2\sin 2x)] dx$  ✓  
 $= \int_0^{\frac{\pi}{2}} [\cos x + 2\sin 2x] dx$   
 $= [\sin x - \cos 2x]_0^{\frac{\pi}{2}}$  ✓  
 $= 1 - (-1) - (0 - 1)$   
 $= 3 \text{ u}^2$  ✓

(b)  $2x^2 + 7x - 3 = 0$

(i)  $\alpha + \beta = -\frac{7}{2}$  ✓

(ii)  $\alpha\beta = -\frac{3}{2}$  ✓

(iii)  $2\alpha^2 + 2\beta^2$   
 $= 2(\alpha^2 + \beta^2)$   
 $= 2[(\alpha + \beta)^2 - 2\alpha\beta]$  ✓  
 $= 2\left[\frac{49}{4} + 3\right]$   
 $= 30.5$  ✓

Q6(a)  
 (i)  $y = -2 \sin x$   
 value  $x = \frac{\pi}{2}$

Q6(c)  $4x^2 + 5x - 3 \equiv A(x-1)^2 + B(x-1) + C$   
 Now  $A = 4$  [coefficient of  $x^2$ ]

Let  $x = 1 \therefore 6 = C$

Let  $x = 0$

$\therefore -3 = A - B + C$

$-3 = 4 - B + 6$

$B = 13$

$\therefore A = 4, B = 13, C = 6$

$4x^2 + 5x - 3 \equiv A(x-1)^2 + B(x-1) + C$   
 $= A(x^2 - 2x + 1) + Bx - B + C$   
 $= Ax^2 + (B - 2A)x + (A - B + C)$

$A = 4$

$B - 2A = 5$

$B - 2(4) = 5$

$B = 13$

$A - B + C = -3$

$4 - 13 + C = -3$

$-9 + C = -3$

$C = 6$

Q6 (c)  $4x^2 + 5x - 3 = A(x-1)^2 + B(x-1) + C$   
 let  $x=1 \therefore 6 = C$   
 let  $x=0$

Q7

a) i)  $8y = x^2 - 2x - 15$

(ii)  $4a = 8$   
 $a = 2$

focus  $(1, 0)$  ✓

$x^2 - 2x + 1 = 8y + 16$  ✓

$(x-1)^2 = 8(y+2)$  ✓

Vertex  $(1, -2)$  ✓

iii)  ~~$y = m(x-1)$~~   
 $y = m(x-1)$

b) i)  $f'(x) = 6x^2 + 6x - 12$

$f''(x) = 12x + 6$

Stationary pts  $f'(x) = 0$   
 $6x^2 + 6x - 12 = 0$

$x^2 + x - 2 = 0$

$(x+2)(x-1) = 0$

$x = -2 \text{ or } 1$  ✓

$f''(-2) = -24 + 6 < 0$

$\therefore$  max at  $(-2, 16)$  ✓

$f''(1) = 12 + 6 > 0$

$\therefore$  min at  $(1, -11)$  ✓

ii)

$f''(x) = 0$

$12x + 6 = 0$

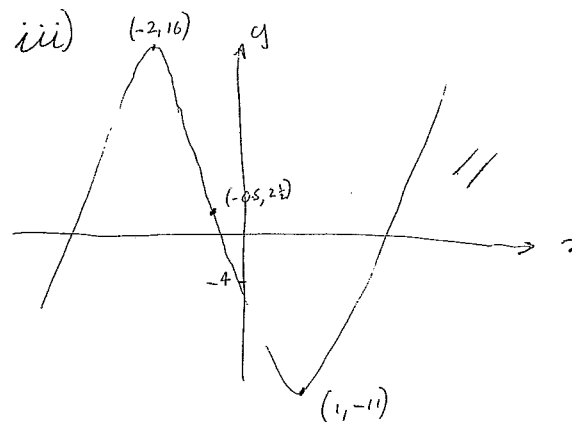
$x = (-\frac{1}{2}, 2\frac{1}{2})$

change of concavity  
 $\therefore$  point of inflection  
 at  $(-0.5, 2\frac{1}{2})$

Check  $x$ 

|       |                |       |
|-------|----------------|-------|
| $-2$  | $-\frac{1}{2}$ | $1$   |
| $y''$ | $< 0$          | $> 0$ |

iii)



iv)  $x > -0.5$

Q8

Q1(a)(i) Consider  $\triangle ADF$  and  $\triangle CBE$ . $AD = CB$  (opposite sides of parallelogram equal) ✓ $\angle ADF = \angle BCE$  (alternate angles  $AD \parallel BC$ )Now  $\angle DAB = \angle BCE$  (opposite angles of parallelogram equal)and since  $AF$  bisects  $\angle DAB$  and  $EC$  bisects  $\angle BCE$ . $\angle DAF = \angle BCE$  $\therefore \triangle ADF \equiv \triangle CBE$  (AAS) ✓(ii)  $DF = EB = 16$  (corresponding sides in congruent triangles)

$$\therefore EF = 40 - 32 = 8 \text{ cm} \quad \checkmark$$

$$(b) (i) v = \frac{dx}{dt} = 25 - \frac{10}{t} \quad \checkmark$$

$$(ii) a = \frac{dv}{dt} = \frac{10}{t^2} \quad \checkmark$$

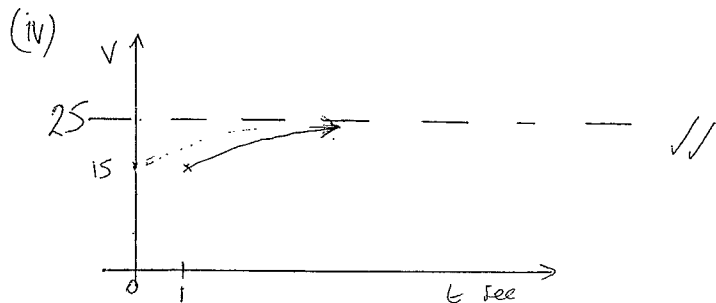
(ii) since  $t$  in seconds find  $t = 60e$  seconds.

$$\therefore v = 25 - \frac{10}{60e}$$

$$= 25 - \frac{1}{6e} \quad \checkmark \quad (24.94)$$

$$a = \frac{10}{(60e)^2}$$

$$= \frac{1}{360e^2} \quad \checkmark \quad (0.00038)$$

(iii) As  $t \rightarrow \infty$   $v \rightarrow 25$  (as  $\frac{10}{t} \rightarrow 0$ ) ✓✓

Q8

Problems with Q1.

(a)(i) On the whole this was done quite poorly.

- some proved similarity using equiangular.
- many tried to show  $AF = EC$  by converse reasoning
- language used was incomplete (eg alternate angles on parallel lines).
- many used given for reasons when inappropriate.

(ii) the majority got this correct, regardless of (i).

(b)(i) Most got these, but a few differentiation issues (eg with negative sign &amp; second term)

(ii) few candidates realised they had to substitute  $t = 60e$  secs.(iii) Many just wrote  $v \rightarrow 25$  with no explanation - two marks usually requires more. Here they needed to show  $\frac{10}{t} \rightarrow 0$ .

(iv) - Many did not attempt.

- Those that did made a number of errors such as
- starting graph at  $t = 0$
- not putting in asymptote

Q9

$$(a) 4 \cos^2 \theta - 3 = 0, 0 \leq \theta \leq 2\pi$$

$$\cos \theta = \pm \frac{\sqrt{3}}{2}$$

$$\therefore \theta = \frac{\pi}{6}, \pi - \frac{\pi}{6}, \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

$$= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6} \quad \checkmark \quad \left[ \begin{array}{l} 3 \text{ for all} \\ 2 \text{ for } \frac{\pi}{6}, \frac{11\pi}{6} \end{array} \right]$$

$$(b) (i) \text{ if } x = y^2, \text{ let } x = 1$$

$$\therefore y = 1 \quad \therefore Q(1, 1) \quad \checkmark \quad (1)$$

$$(ii) \text{ Volume of Solid} = \pi \int_1^4 x^2 dy - \text{Volume of cylinder}$$

$$= \pi \int_1^4 y^4 dy - \pi \times 1^2 \times 3$$

$$= \pi \left[ \frac{y^5}{5} \right]_1^4 - 3\pi$$

$$= \frac{\pi}{5} \times 1023 - 3\pi$$

$$= \frac{1008}{5} \pi \approx 633.365 \quad \checkmark \quad \text{ACCEPT}$$

$$(c) (i) \text{ If } Q = Q_0 e^{-kt}$$

$$\frac{dQ}{dt} = -k Q_0 e^{-kt}$$

$$= -kQ \quad \checkmark$$

$$(ii) 0.15 Q_0 = Q_0 e^{-kt} \quad \checkmark$$

$$(ii) \therefore \frac{Q_0}{2} = Q_0 e^{-4500k}$$

$$\therefore e^{-4500k} = \frac{1}{2}$$

$$-4500k = \ln \frac{1}{2}$$

$$k = -\frac{1}{4500} \ln \frac{1}{2}$$

$$= \frac{1}{4500} \ln 2 \quad \checkmark$$

$$e^{-kt} = 0.15$$

$$-kt = \ln 0.15$$

$$t = \frac{\ln 0.15}{-k}$$

$$= 17316.35 \text{ year}$$

$$= 12000 \text{ years} \quad \checkmark$$

Q10

$$(a) (i) A(x) = x \times y = x e^{-2x} \quad \checkmark$$

$$(ii) A'(x) = -2x e^{-2x} + e^{-2x}$$

$$= e^{-2x} (1 - 2x) \quad \checkmark$$

$$\text{let } A'(x) = 0 \therefore x = \frac{1}{2}$$

Test for MAX/MIN using first derivative

|           |               |     |
|-----------|---------------|-----|
| x: 0.4    | $\frac{1}{2}$ | 0.6 |
| A(x): > 0 | 0             | < 0 |

MAX

$$\therefore \text{as } x = \frac{1}{2}$$

$$y = e^{-1}$$

$$\therefore A\left(\frac{1}{2}, \frac{1}{e}\right)$$

$$(b) (i) \text{ If rate} = 9\% \text{ pa}$$

$$\text{rate/month} = 0.75\%$$

$$\therefore \text{Amount owing after 1 month} = P(1.0075) - 2700$$

$$\checkmark \text{ after 2 months} = P(1.0075)^2 - 2700(1.0075)$$

$$\checkmark \text{ after 3 months} = P(1.0075)^3 - 2700(1.0075)^2 - 2700$$

$$= P(1.0075)^3 - 2700(1 + 1.0075 + 1.0075^2)$$

$$(ii) \therefore A_n = P(1.0075)^n - \frac{2700(1.0075^n - 1)}{1.0075 - 1}$$

$$\text{let } A_n = 0 \therefore P = \frac{360000(1.0075^n - 1)}{1.0075^n}$$

Q10

(b) (iii) let  $n = 360$  ✓

$$\therefore P = \frac{360000(1.0075^{360} - 1)}{1.0075^{360}} \quad (2)$$

$$= \$335561.04 \quad \checkmark$$

$$\begin{aligned} \text{(iv) Interest paid} &= 360 \times 2700 - 335561.04 \\ &= \$636438.96 \quad \checkmark \end{aligned} \quad (1)$$